of Comp

National University of Computer and Emerging Sciences
Lahore Campus

DLD (EE1005)

Final Exam

Date: May 20^{th} 2024

Course Instructor(s)

Mr. Kamran Jamal

Ms. Tamania Javaid

Total Time (Hrs): 3

Total Marks: 100

Total Questions: 4

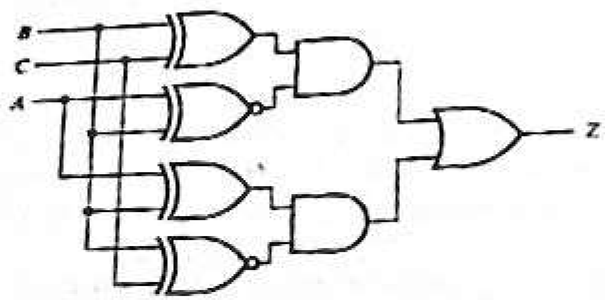
Roll No ________ Section ________ Student Signature ________

Attempt all the questions. All questions carry equal marks and equal weights.

CLO #2: Apply Boolean Algebra and K-map methods to optimize logic circuits

Q1:

Consider the following circuit



a) Construct the truth table for the function Z.

b) Write the function Z in canonical sum of product form (little m notation).

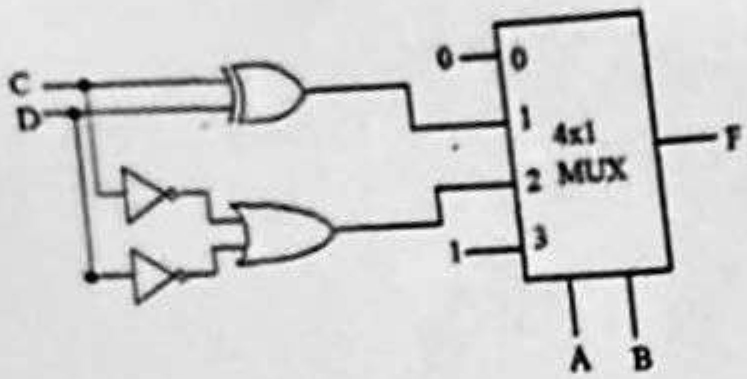
c) Simplify the function to the minimum number of literals.

CLO #3: Design combinational circuits using functional blocks

Q2:

a) Design a combinational multiplier that has two 2-bit inputs A_1A_0 and B_1B_0 and 4 bit output $F_3F_2F_1F_0$. Construct the truth table and implement it using 4 to 16 line decoder and gates of your choice.

b) **Construct** the Truth table and write F in $\sum m$ notation.



CLO #4: Analyze elements of sequential circuits and design finite state machines

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Q3:

a) **Analyze** a sequential circuit with three D flip-flops (A, B, C) defined by the following next state equations

$$D_A = B\overline{C}$$
$$D_B = C$$
$$D_C = \overline{B} + X$$

Where A, B and C are the outputs of the flip-flops A, B and C respectively, X is the external input and D_A, D_B and D_C are the input of flip-flops A, B and C respectively. Construct the state table and state diagram of this sequential circuit.

b) Consider a sequence recognizer circuit that has a serial input S and one output Z. The output becomes one when a sequence 1001 is detected and zero otherwise. **Generate** a state diagram only. Circuit implementation is not required.

CLO #5: Design of sequential circuits for various applications

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Q4:

Design a synchronous binary arbitrary counter using T Flip-Flop to produce the sequence 1,4,7,3,1 and repeat.

a) **Construct** the state diagram and state table for the requied circuit.
b) **Construct** the logic diagram of your design using T- flip flops and logic gates.
c) Include the unused states in the state diagram and explain if this counter has self correction capability.